

Image Segmentation - 3

Snakes: Active Contour Models

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Abstract

A snake is an energy-minimizing spline guided by external constraint forces and influenced by image forces that pull it toward features such as lines and edges. Snakes are active contour models: they lock onto nearby edges, localizing them accurately. Scale-space continuation can be used to enlarge the capture region surrounding a feature. Snakes provide a unified account of a number of visual problems, including detection of edges, lines, and subjective contours; motion tracking; and stereo matching. We have used snakes successfully for interactive interpretation, in which user-imposed constraint forces guide the snake near features of interest.

Snakes: Active Contour Models

1. The position of a 2D contour can be parameterized as

$$\vec{V}(s) = (x(s), y(s))^T$$

where s is a parameter that increases as the contour is traversed, $0 \leq s \leq 1$, and x and y are position variables as well as functions of s . It can be either an open or close contour. Link: <http://www.cs.ucla.edu/~dt/papers/ijcv88/ijcv88.pdf>

2. Snakes are an attractive approach because they are capable of finding salient image contours – edges, lines and subjective contours – as well as tracking those contours during motion.

Demo: <http://www.markschulze.net/snakes/>

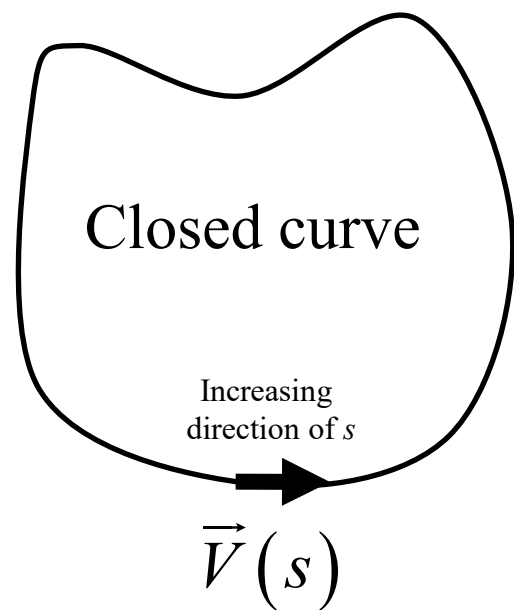
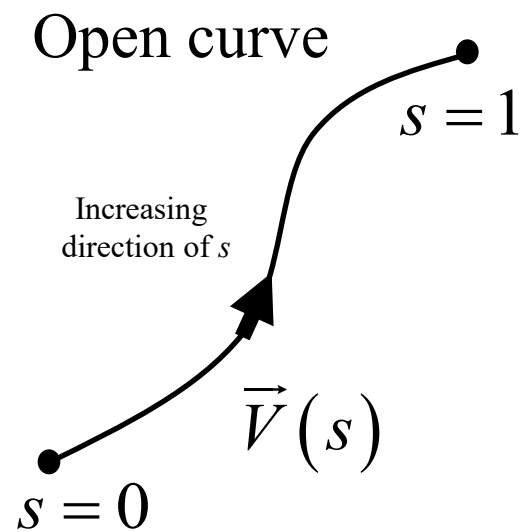
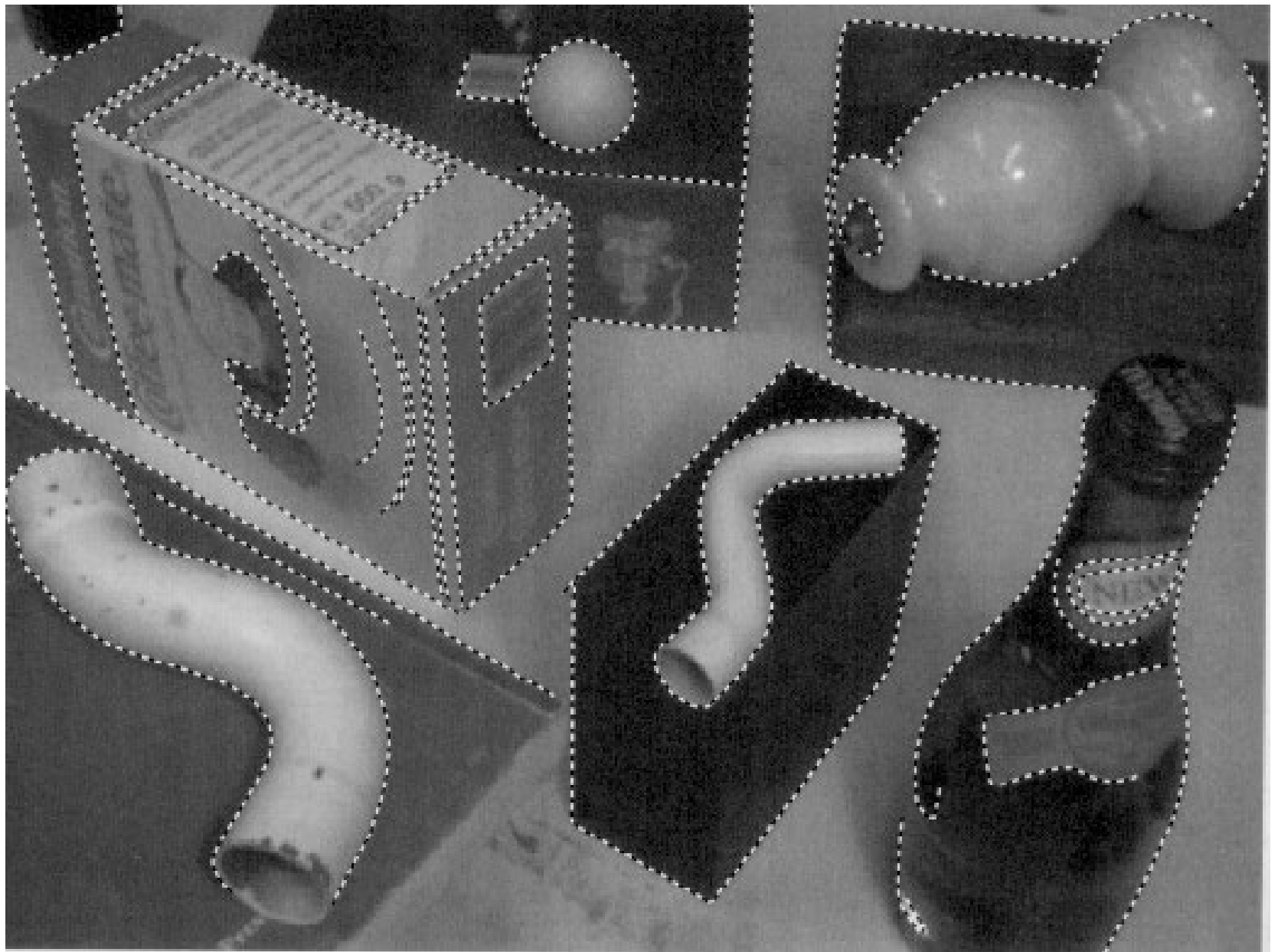


FIGURE 1 Snake (white) attracted to cell membrane in an EM photomicrograph [18].



Examples

Snakes: Active Contour Model



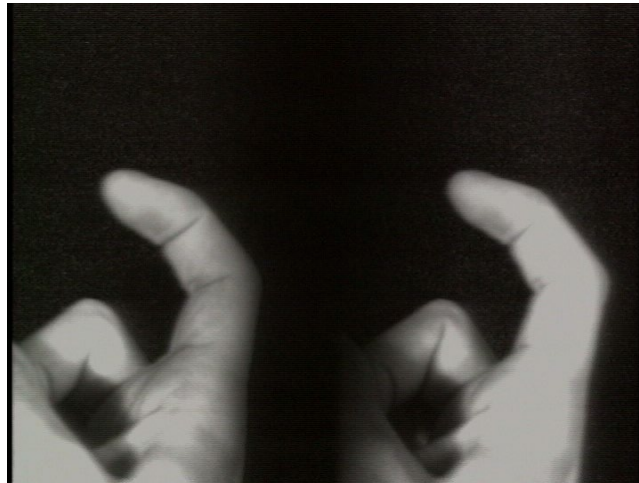
Tracking



Source: <http://www.cs.ucla.edu/~dt/vision.html>

Example

Finger tracking



Source: <http://www.cs.ucla.edu/~dt/vision.html>

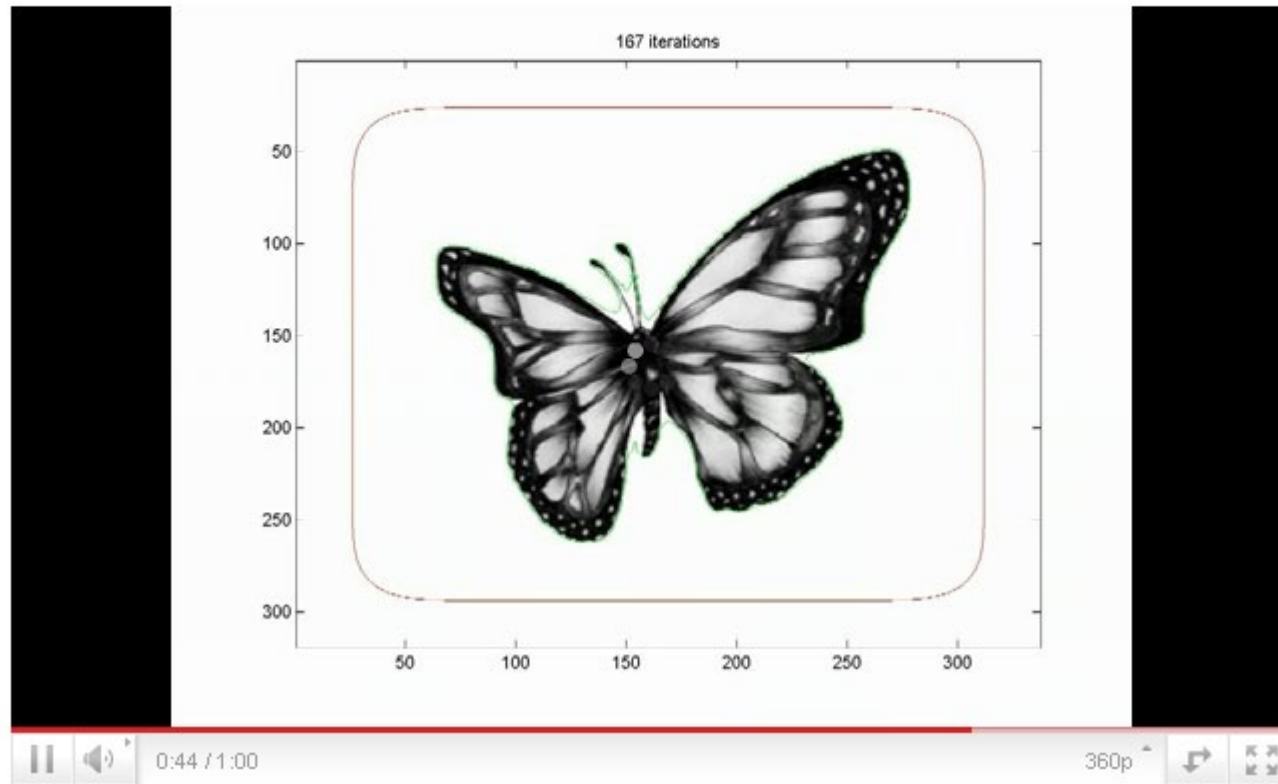
 Search

Car Tracking with Active Contours

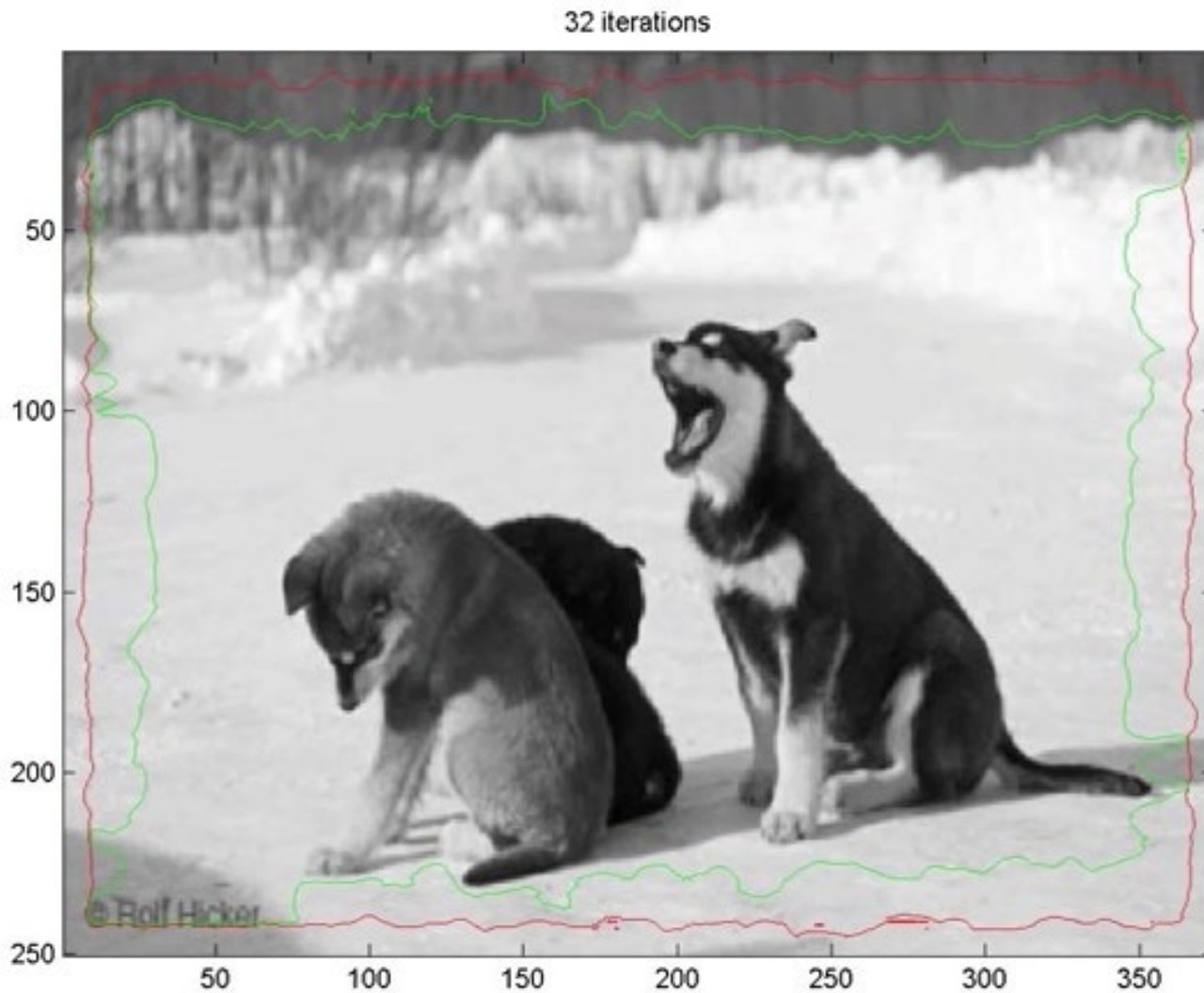


Source: <http://www.youtube.com/watch?v=5se69vcbqxA&feature=related>

Geometric Active Contour evolution for a Butterfly Image

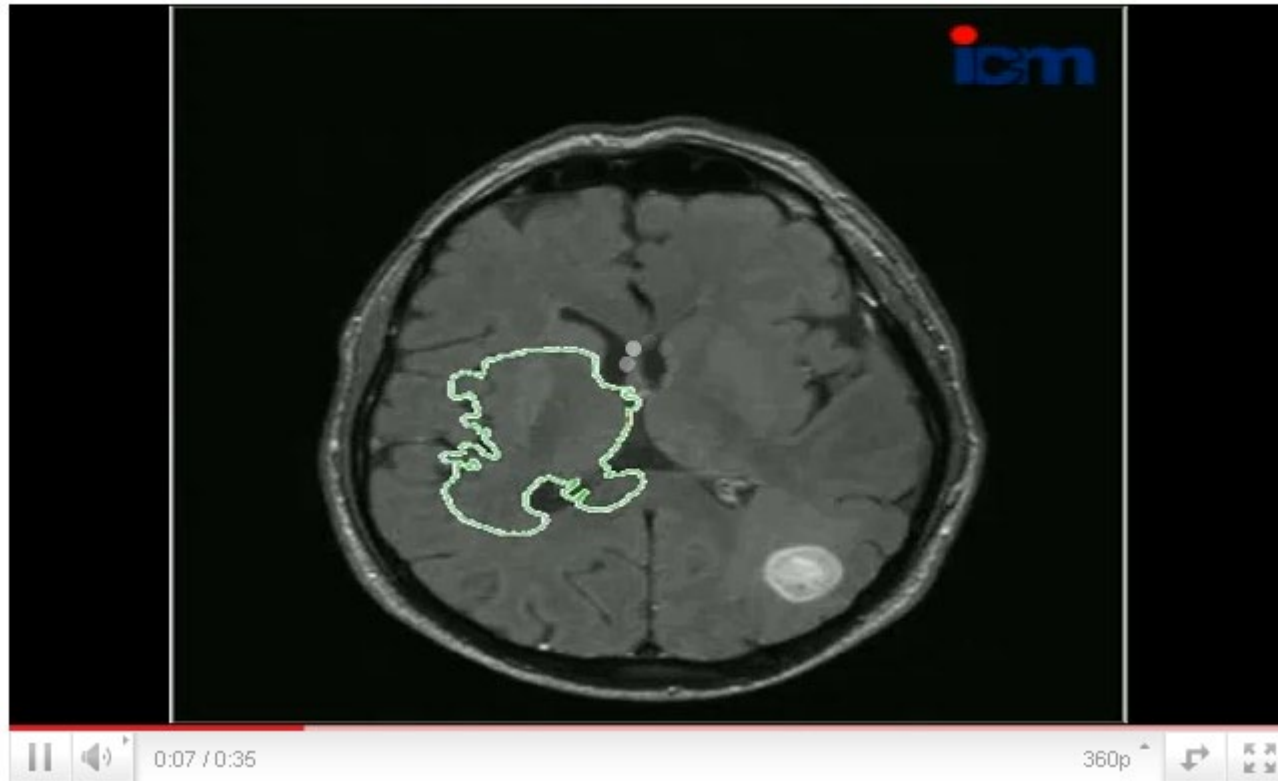


Source: <https://www.youtube.com/watch?v=qlaJMiARUyg>



Source: https://www.youtube.com/watch?v=QNk6Zx6Wi_k

Geometrical image segmentation



Source: http://www.youtube.com/watch?v=3imS_9EeNhU&feature=related

Snakes: Active Contour Models

3. The shape of the contour is dictated by the energy functional

$$E(\vec{V}) = E_{\text{int}}(\vec{V}) + E_{\text{ext}}(\vec{V})$$

4. Internal energy for controlling the intrinsic contour behavior

$$E_{\text{int}}(\vec{V}) = \frac{1}{2} \int_0^1 \alpha(s) \left| \frac{d\vec{V}}{ds} \right|^2 ds + \frac{1}{2} \int_0^1 \beta(s) \left| \frac{d^2\vec{V}}{ds^2} \right|^2 ds$$

- a. the first term controls the *tension / elasticity / stretching* of the contour.
- b. the second term controls the *rigidity / stiffness / bending / curvature* of the contour.

Snakes: Active Contour Models

5. External energy for encouraging contour onto image structures

$$E_{\text{ext}}(\vec{V}) = \int_0^1 E_{\text{image}}(\vec{V}(s)) ds$$

- a. E_{image} represents the scalar potential (gradient) function defined on the image plane, e.g.,

$$E_{\text{image}}(\vec{V}(s)) = -c \left| \nabla \left[G_{\sigma} * I(\vec{V}(s)) \right] \right|$$

- b. $c > 0$ is constant, $G_{\sigma} * I$ represents an image I convolved with a Gaussian smoothing filter with standard deviation, $\text{SD} = \sigma$.

6. The final shape of the contour $\vec{V}(s)$ corresponds to the minimum of energy E .

Examples

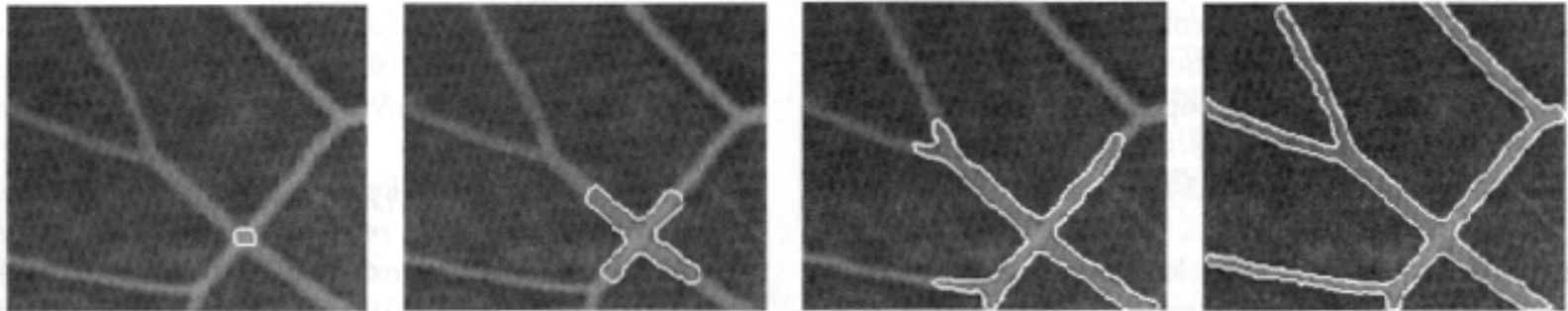


FIGURE 4 Image sequence of clipped angiogram of retina showing an automatically subdividing snake flowing and branching along a vessel [96].

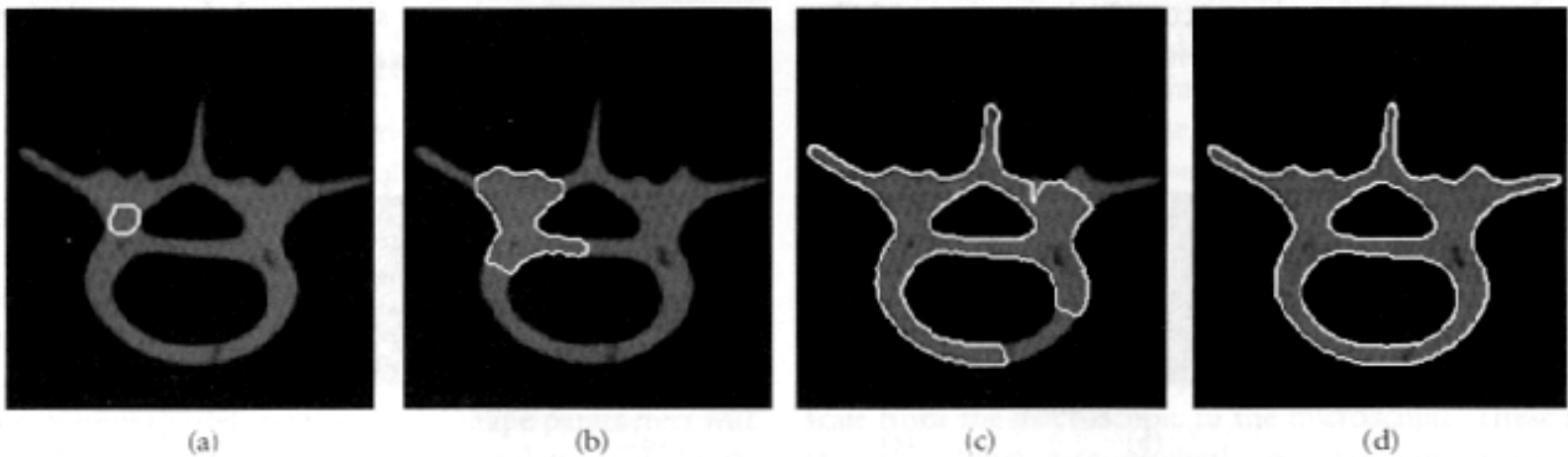
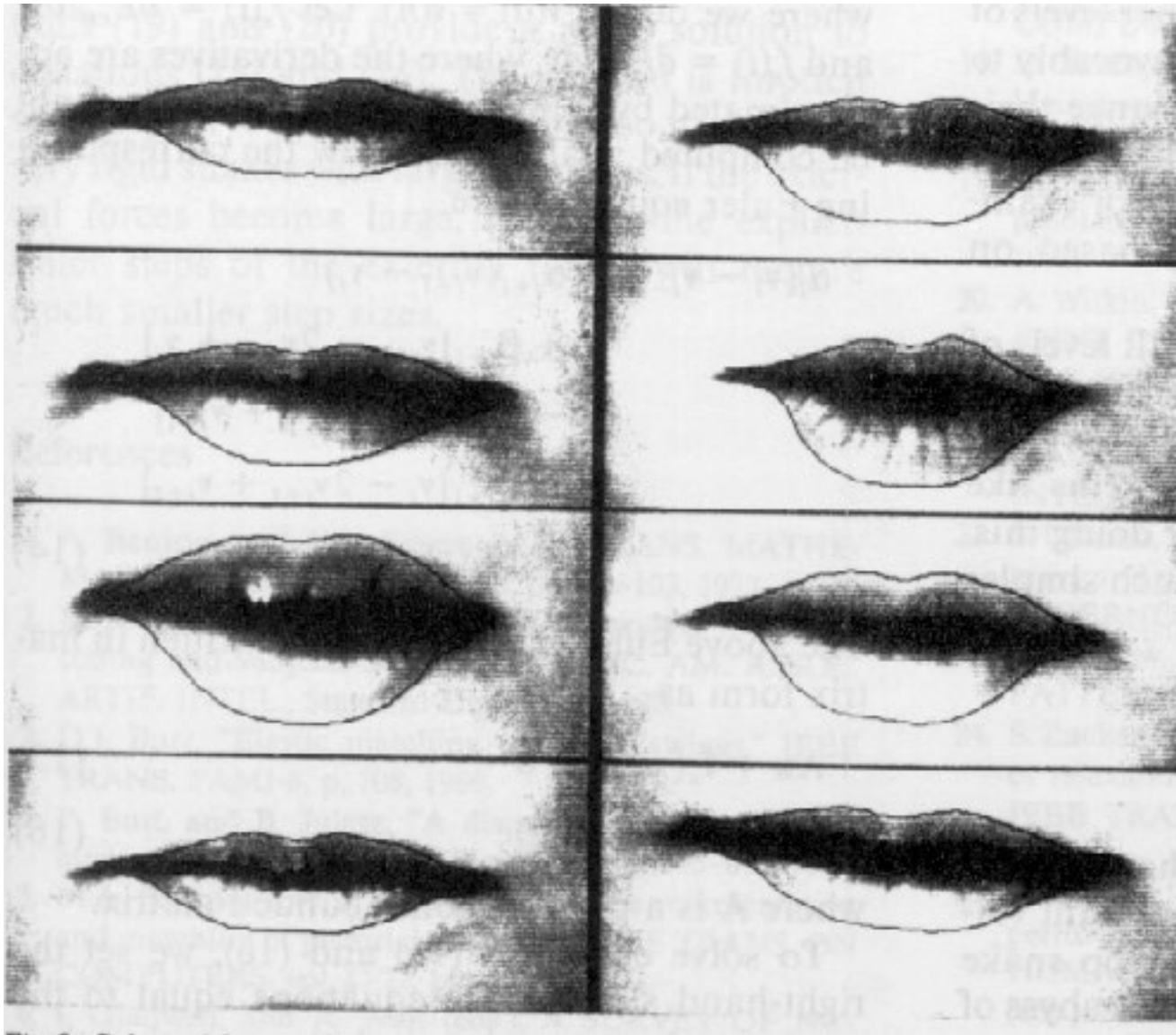


FIGURE 5 Segmentation of a cross sectional image of a human vertebra phantom with a topologically adaptable snake [96]. The snake begins as a single closed curve and becomes three closed curves.

Example



Kass, Witkin and
Terzopoulos, 1987

Region Based Segmentation Using Active Contours

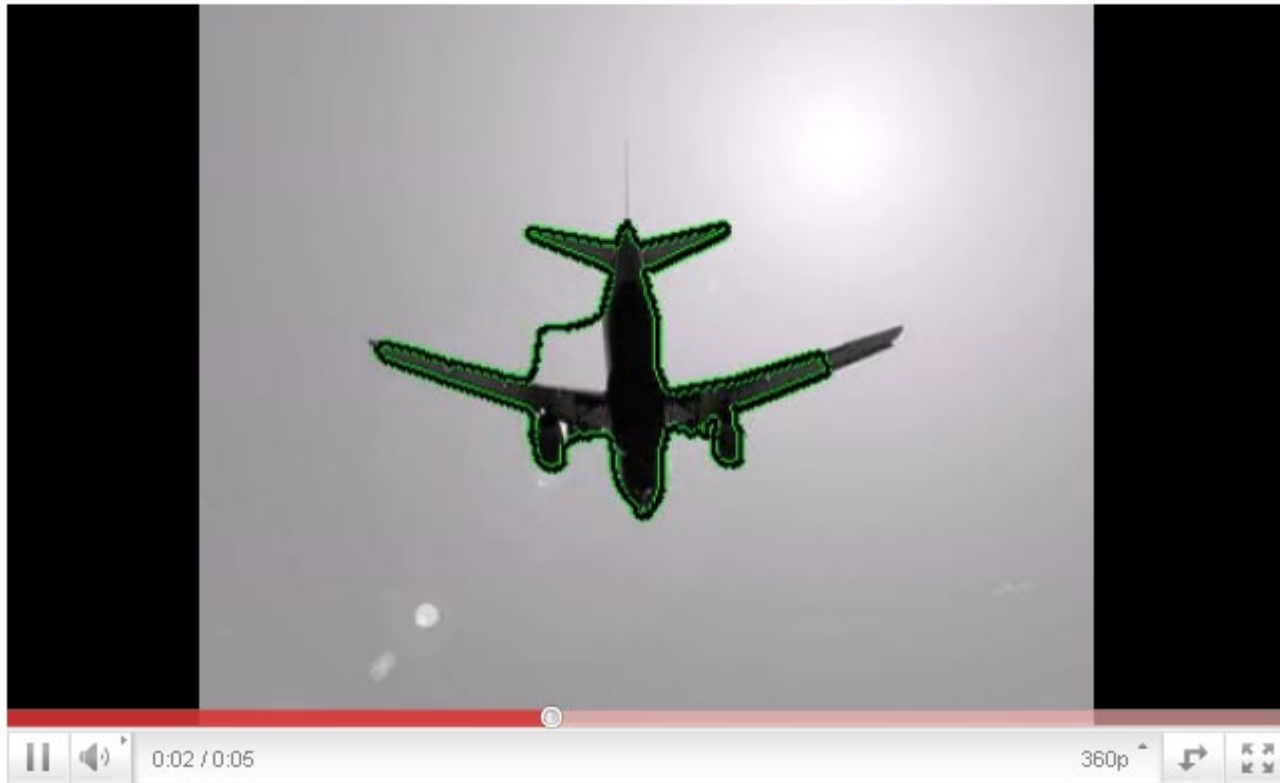


<https://www.youtube.com/watch?v=ceIddPk78yA>



Search

Chan Vese Segmentation



<https://www.youtube.com/watch?v=H3P5N7ZfvEo>

Snakes: Active Contour Models

7. According to the calculus of variations, the contour $\vec{V}$ which minimizes the energy E must satisfy the Euler-Lagrange equation,

$$-\frac{d}{ds}\left(\alpha(s)\frac{d\vec{V}}{ds}\right) + \frac{d^2}{ds^2}\left(\beta(s)\frac{d^2\vec{V}}{ds^2}\right) + \nabla E_{\text{ext}}(\vec{V}) = 0$$

where $\nabla E_{\text{ext}}(\vec{V}) = \begin{pmatrix} \frac{\partial E_{\text{ext}}(\vec{V})}{\partial x} \\ \frac{\partial E_{\text{ext}}(\vec{V})}{\partial y} \end{pmatrix}$

Each term generates a “force” that drives the contour deformation. At min energy, the forces are balanced.

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8. Discrete formulation based on finite differences

$$-\frac{d}{ds}\left(\alpha(s)\frac{d\vec{V}}{ds}\right)+\frac{d^2}{ds^2}\left(\beta(s)\frac{d^2\vec{V}}{ds^2}\right)+\nabla E_{\text{ext}}(\vec{V})=0$$

Using the finite difference scheme in space with a step size of h

$$\begin{aligned} &\frac{1}{h}\left(a_i(\vec{V}_i - \vec{V}_{i-1}) - a_{i+1}(\vec{V}_{i+1} - \vec{V}_i)\right) + \\ &\frac{b_{i-1}}{h^2}(\vec{V}_{i-2} - 2\vec{V}_{i-1} + \vec{V}_i) - 2\frac{b_i}{h^2}(\vec{V}_{i-1} - 2\vec{V}_i + \vec{V}_{i+1}) + \frac{b_{i+1}}{h^2}(\vec{V}_i - 2\vec{V}_{i+1} + \vec{V}_{i+2}) - \\ &\left(f_x(\vec{V}_i), f_y(\vec{V}_i)\right)^T = 0 \end{aligned}$$

where h = step size

$$\vec{V}_i = (x(ih), y(ih))^T$$

$$a_i = \frac{\alpha(ih)}{h}$$

$$b_i = \frac{\beta(ih)}{h^2}$$

$$f_x(\vec{V}_i) = \frac{\partial E_{\text{ext}}}{\partial x} \quad \text{and} \quad f_y(\vec{V}_i) = \frac{\partial E_{\text{ext}}}{\partial y}$$

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9. Matrix form

$$\mathbf{A}\vec{x} + \vec{f}_x = 0 \quad \text{and} \quad \mathbf{A}\vec{y} + \vec{f}_y = 0$$

where $\mathbf{A}$ is a matrix in terms of a and b .

$$f_x(\vec{V}_i) = \frac{\partial E_{\text{ext}}}{\partial x} \quad \text{and} \quad f_y(\vec{V}_i) = \frac{\partial E_{\text{ext}}}{\partial y}$$

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10. Dynamic deformable model. Let $\vec{V} = \vec{V}(s, t)$ be a time-varying contour. The evolution equation is given as

$$\frac{\partial \vec{V}}{\partial t} - \frac{\partial}{\partial s} \left(\alpha \frac{\partial \vec{V}}{\partial s} \right) + \frac{\partial^2}{\partial s^2} \left(\beta \frac{\partial^2 \vec{V}}{\partial s^2} \right) + \nabla E_{\text{ext}}(\vec{V}) = 0$$

11. Discrete formulation of the dynamic SNAKE (Finite difference) in matrix form

$$\gamma(\vec{x}_t - \vec{x}_{t-1}) + A\vec{x}_t + \vec{f}_x(x_{t-1}, y_{t-1}) = 0$$

$$\gamma(\vec{y}_t - \vec{y}_{t-1}) + A\vec{y}_t + \vec{f}_y(x_{t-1}, y_{t-1}) = 0$$

where γ represents time step size.

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12. Therefore, the dynamic SNAKE can be solved iteratively.

$$\vec{x}_t = (A + \gamma I)^{-1} \left(\gamma \vec{x}_{t-1} - \vec{f}_x(x_{t-1}, y_{t-1}) \right)$$
$$\vec{y}_t = (A + \gamma I)^{-1} \left(\gamma \vec{y}_{t-1} - \vec{f}_y(x_{t-1}, y_{t-1}) \right)$$

13. At equilibrium, a stationary contour with minimum internal and external energies is obtained.

Quadratic energy functional, example 1

$$E_{\text{ext}} = \frac{-1}{2}(u - v)^2$$

where $u =$ mean in region R_u

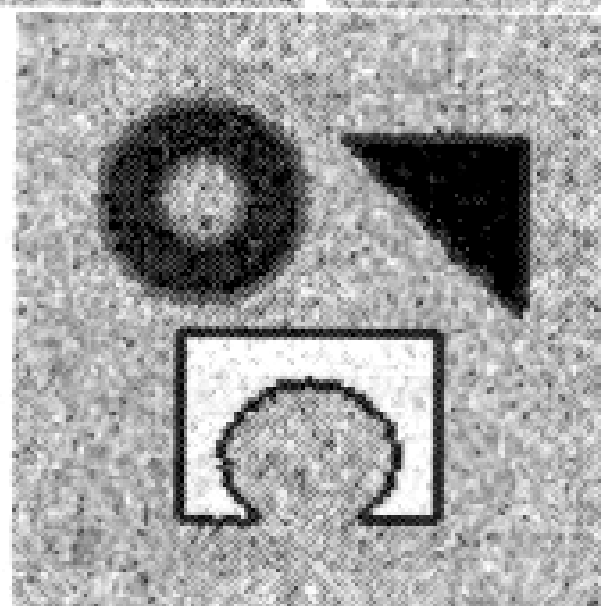
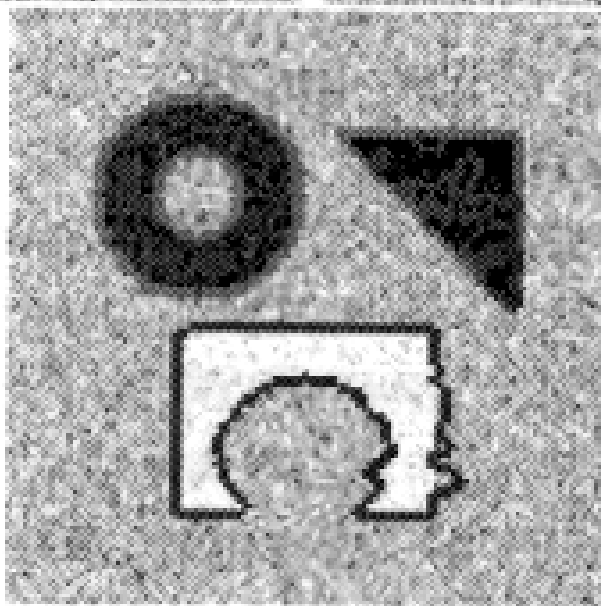
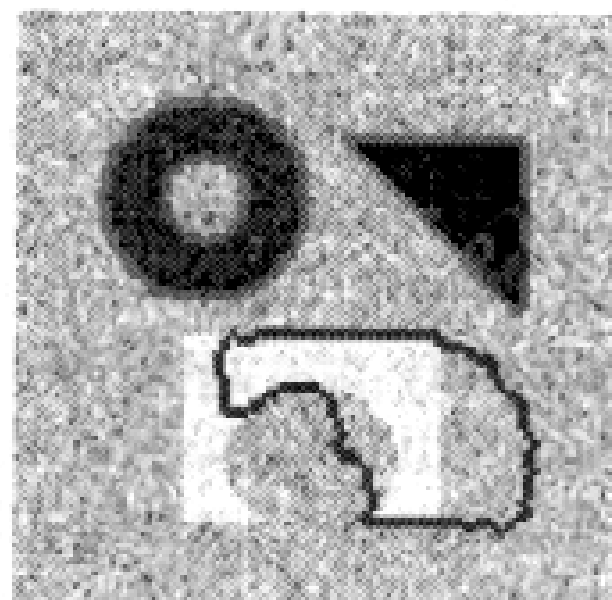
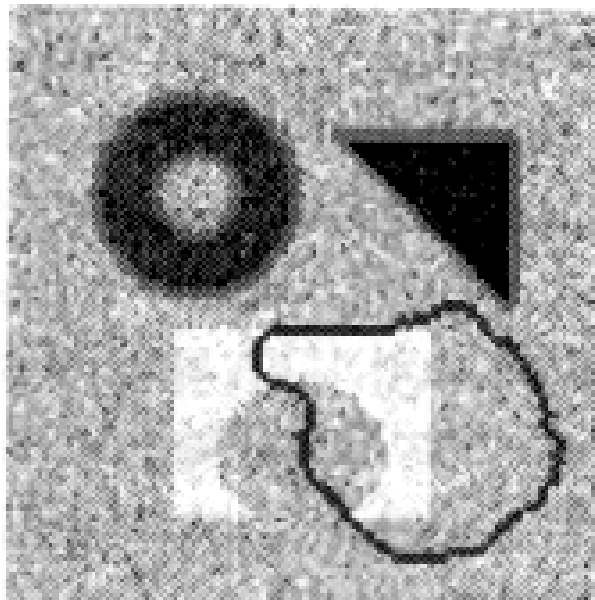
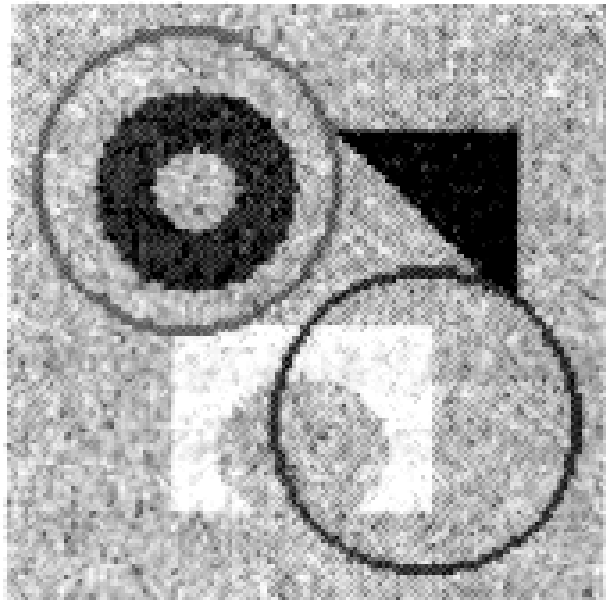
$$A_u = \int_{R_u} dA = \text{area in region } R_u$$

$v =$ mean in region R_v

$$A_v = \int_{R_v} dA = \text{area in region } R_v$$

$$\begin{aligned} \frac{d\vec{V}}{dt} &= -\nabla E_{\text{ext}} \\ &= (u - v) \left(\frac{I - u}{A_u} + \frac{I - v}{A_v} \right) \vec{n} \quad \vec{n} = \text{outward normal} \end{aligned}$$

Example 1



Active Contours Without Edges

Tony F. Chan, *Member, IEEE*, and Luminita A. Vese

Abstract—In this paper, we propose a new model for active contours to detect objects in a given image, based on techniques of curve evolution, Mumford–Shah functional for segmentation and level sets. Our model can detect objects whose boundaries are not necessarily defined by gradient. We minimize an energy which can be seen as a particular case of the minimal partition problem. In the level set formulation, the problem becomes a “mean-curvature flow”-like evolving the active contour, which will stop on the desired boundary. However, the stopping term does not depend on the gradient of the image, as in the classical active contour models, but is instead related to a particular segmentation of the image. We will give a numerical algorithm using finite differences. Finally, we will present various experimental results and in particular some examples for which the classical snakes methods based on the gradient are not applicable. Also, the initial curve can be anywhere in the image, and interior contours are automatically detected.

the image (the external energy). Observe that, by minimizing the energy (1), we are trying to locate the curve at the points of maxima $|\nabla u_0|$, acting as an edge-detector, while keeping a smoothness in the curve (object boundary).

A general edge-detector can be defined by a positive and decreasing function g , depending on the gradient of the image u_0 , such that

$$\lim_{z \rightarrow \infty} g(z) = 0.$$

For instance

$$g(|\nabla u_0(x, y)|) = \frac{1}{1 + |\nabla G_\sigma(x, y) * u_0(x, y)|^p}, \quad p \geq 1$$

Quadratic energy functional, example 2

$$E_{\text{ext}} = \int_{\Omega_1} |I(x, y) - C_1|^2 dx dy + \int_{\Omega_2} |I(x, y) - C_2|^2 dx dy$$

$$\frac{d\vec{V}}{dt} = (|I(x, y) - C_2|^2 - |I(x, y) - C_1|^2) \vec{n}$$

$\vec{n}$ = outward normal

Example 2

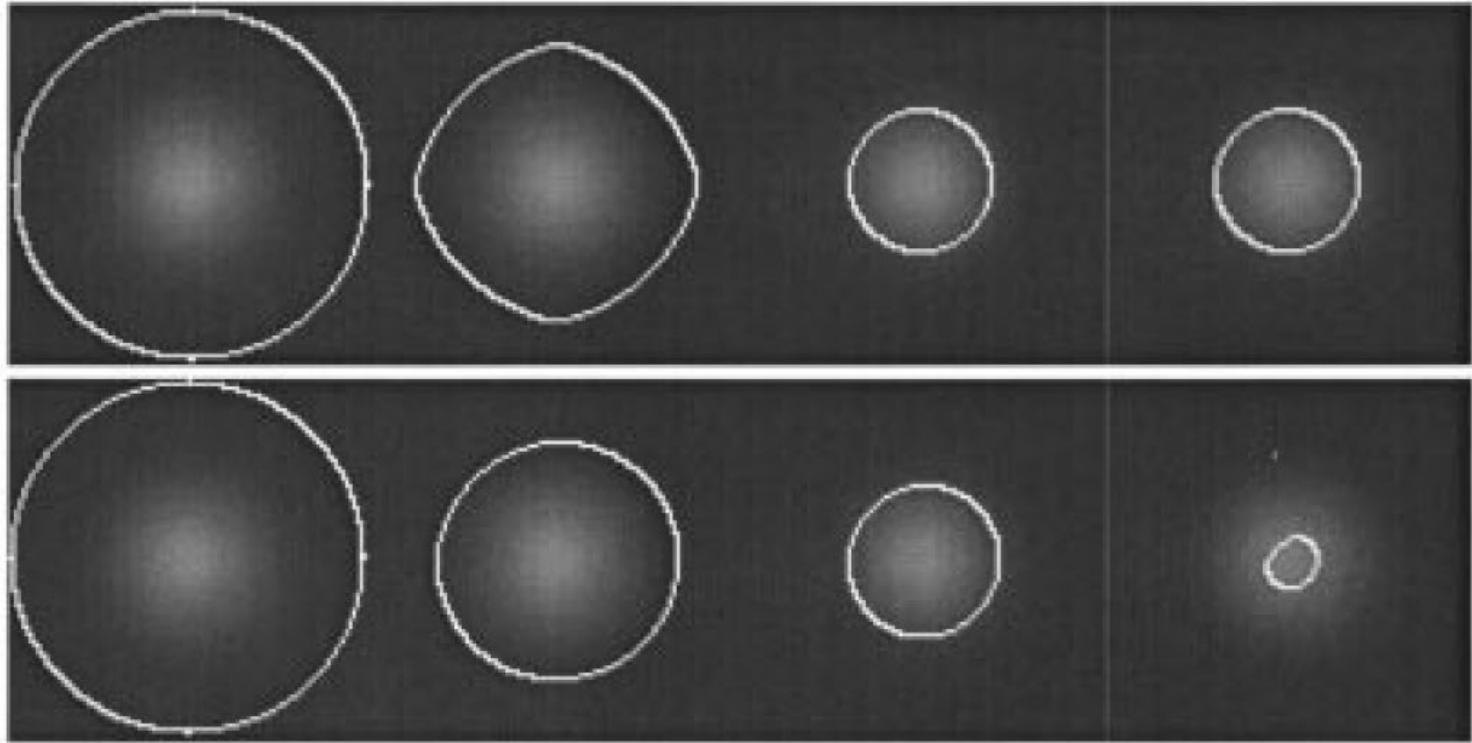
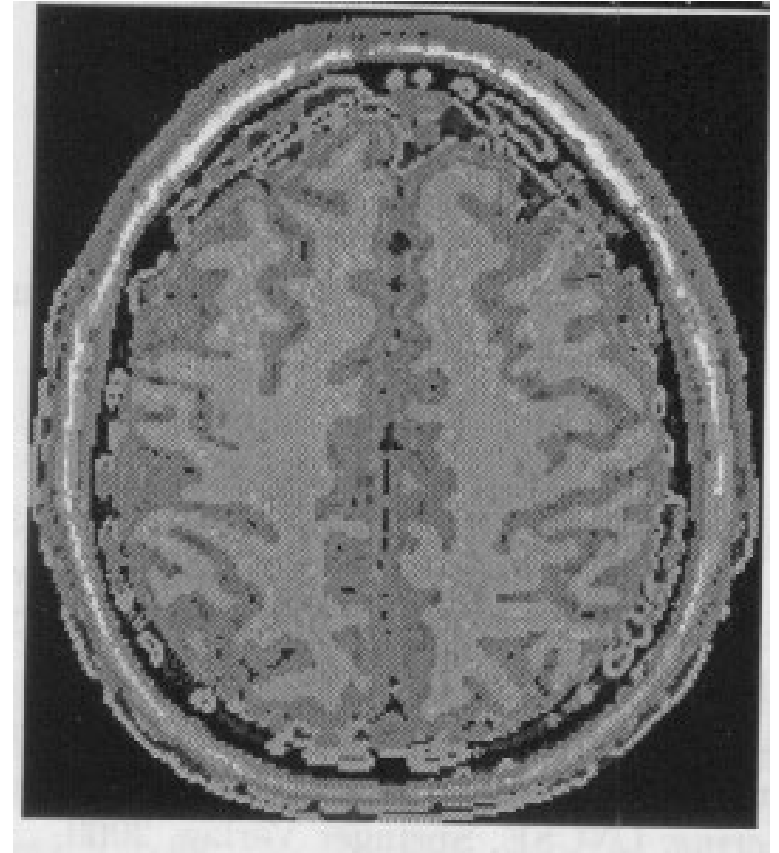
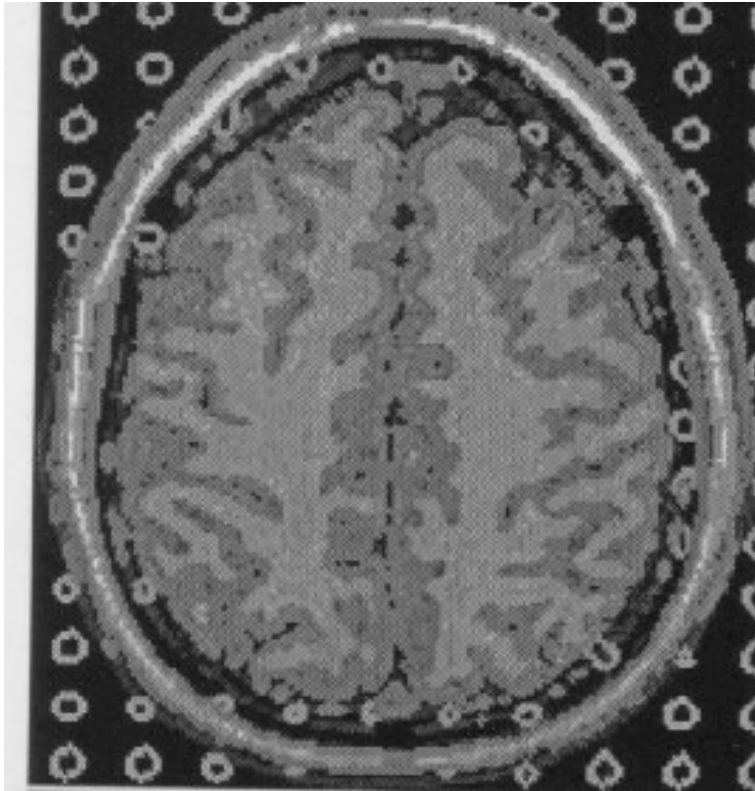


Fig. 9. Object with smooth contour. Top: results using our model without edge-function. Bottom: results using the classical model (2) with edge-function.

Example using level set methods



Example using level set methods

